

BLOWFISH ALGORITHM

AIM

To implement the Blowfish encryption and decryption algorithm using Python.

Algorithm

1. Start the program.
2. Import the Blowfish cipher library.
3. Enter the plaintext.
4. Enter a secret key.
5. Pad the plaintext to make it a multiple of the block size.
6. Encrypt the plaintext using Blowfish.
7. Display the ciphertext.
8. Decrypt the ciphertext.
9. Display the original plaintext.
10. Stop.

Python Code

```
from Crypto.Cipher import Blowfish
from Crypto.Util.Padding import pad, unpad
import base64

plaintext = input("Enter the Plain Text: ")
key = input("Enter the Secret Key: ").encode()
cipher = Blowfish.new(key, Blowfish.MODE_ECB)
ciphertext = cipher.encrypt(pad(plaintext.encode(), Blowfish.block_size))
print("\nEncrypted Text:")
print(base64.b64encode(ciphertext).decode())

decrypted = unpad(cipher.decrypt(ciphertext), Blowfish.block_size)
print("\nDecrypted Text:")
print(decrypted.decode())
```

Output

```
... Enter the Plain Text: hello world
Enter the Secret Key: security

Encrypted Text:
J4zx84nhTe5bowB2iF7htw==

Decrypted Text:
hello world
```

The result:

The above code is executed and the output is obtained successfully.